



Ben-Gurion University of the Negev
The Faculty of Natural Sciences
The Department of Computer Science

Expected-Return Inference over Complete Policies

Thesis submitted in partial fulfillment of the requirements
for the Master of Sciences degree

Eden Havri

Under the supervision of **Ronen Brafman**

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Abstract

We study generalized Bayesian inference over complete deterministic policies using the expected return of a stochastic finite-horizon Markov decision process. The Gibbs target is defined before choosing an approximation: environment randomness is averaged before the value is exponentiated. This established policy-search formulation is distinct from weighting individual trajectories by exponentiated realized return.

The current practical method learns a factorized distribution over complete policy tables. A separately costed PPO guide initializes the distribution; direct variational refinement then uses simulator returns, a leave-one-out baseline, and an exact policy-space entropy term. We justify its visited-state-masked score gradient and verify the actual implementation against exact differentiation on enumerable instances. One rollout per sampled policy is

sufficient for gradient correctness, although allocation between policy draws and repeated rollouts affects finite-budget learning. The method samples fresh policies between updates and commits to one table during each evaluation episode. It has no particle-resampling stage.

The empirical design separates gradient correctness, variational family error, and control performance. Strict interaction caps include guide training and refinement; native stochastic execution and committed-policy execution are both reported. Preliminary GridWorld studies favor one rollout per policy in a medium-map allocation test and find similar values to exploration-assisted PPO in one matched-budget comparison. A failed hard run exposes insufficient exploration; reallocating the same budget toward guide training improves its subsequent control without eliminating the gap between execution rules. These are development results, not confirmatory superiority or general posterior-fidelity claims.

The thesis also documents exact-score and fixed-bank policy MCMC, replicated SMC, their validation contracts, and unsuccessful development configurations. The planned larger comparison includes REINFORCE, PPO-Clip, discrete SAC, categorical cross-entropy search, and Double Q-learning. Independent guide seeds, held-out maps, family-error analysis, and the complete cost–performance comparison remain necessary before final conclusions.

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Chapter 1

Introduction

Planning in a stochastic environment requires two conceptually separate operations. First, a policy is evaluated by averaging its return over the environment's transition randomness. Second, a decision rule selects among policies. Conflating these operations changes the objective: exponentiating a single realized return before averaging is generally not equivalent to exponentiating the expected return.

This manuscript develops a policy-inference formulation in which the target is defined before an approximation algorithm is chosen. The target assigns mass to complete deterministic policies according to expected return. Its modes are therefore expected-return maximizers, while a finite inverse temperature retains a distribution over alternatives. The construction is a generalized Gibbs posterior over decisions, not a posterior belief about unknown transition parameters. Related free-utility formulations also trade expected utility against information cost [1], and recent general Bayesian policy learning

updates distributions over decision rules from a loss [2]. Our setting differs by targeting finite-horizon MDP policies and by making the expected-return order of operations explicit.

Control-as-inference methods provide a broad connection between optimal control and probabilistic inference [3]. In stochastic dynamics, however, the choice of graphical model and optimality likelihood determines whether the resulting objective is risk-neutral, risk-sensitive, or maximum-entropy. We therefore do not treat “inference” as sufficient to identify the control objective. Expected-return Gibbs policy search and Metropolis–Hastings inference over policies have direct precedent in Wingate et al. [4], who also discuss replica exchange. Neither the exponential target, its mode preservation, nor policy-space MCMC is claimed here as a new inference principle. The research question is how to approximate this target at a defensible computational cost, and what control behavior follows from committing to a sampled policy. The current practical candidate uses direct variational learning; corrected MCMC and fixed-bank SMC provide separately assessed alternatives. Preliminary development comparisons motivate, but do not replace, held-out validation.

The study has five components:

1. It specifies an expected-return Gibbs measure over complete policies, including its reference measure, temperature, and execution semantics.
2. It gives exact finite-policy oracles and variance-sensitive diagnostics that detect substitution of an exponentiated-trajectory objective.
3. It implements direct policy-space variational learning with an unbiased masked score-function gradient, independent baseline, and exact policy-

table entropy term.

4. It separates variational family error, fixed-tape approximation, and MCMC mixing, retaining corrected guided MCMC and replicated SMC as diagnostic alternatives rather than conflating their guarantees.
5. It defines a fail-fast comparison against five published algorithmic families with both policy-commitment and native execution semantics, strict total interaction caps, and separately labeled exploration aids.

Exact small-instance checks validate target definitions, gradient expectations, and sampling kernels before scaling. They do not establish full posterior agreement for the variational family. The protocol is specified before confirmatory results; this draft labels development findings as preliminary.

Chapter 2

Problem Setting and Notation

2.1 Finite-horizon stochastic control

Let an episodic Markov decision process be $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \rho_0, H, \gamma)$, where $T(\cdot \mid s, a)$ is a transition kernel, $R(s, a, s')$ is a bounded reward, ρ_0 is the initial-state distribution, and H is the rollout horizon. The discount is $\gamma \in (0, 1]$; the current GridWorld study uses $\gamma = 1$.

For a deterministic policy $\pi \in \Pi$, a trajectory $\tau = (s_1, a_1, \dots, s_{H+1})$ is generated by

$$s_1 \sim \rho_0, \quad a_t = \pi(s_t), \quad s_{t+1} \sim T(\cdot \mid s_t, a_t).$$

Its truncated return and the policy value are

$$G_H(\tau) = \sum_{t=1}^H \gamma^{t-1} R(s_t, a_t, s_{t+1}), \quad J_H(\pi) = \mathbb{E}_{\tau \sim p(\cdot \mid \pi)}[G_H(\tau)]. \quad (2.1)$$

The implemented policy class is stationary in the physical state. Time-to-go is deliberately not appended to the observation, so recurrent states at different remaining horizons can be aliased. Every learned and inferred method shares this restriction. Finite-horizon dynamic programming is reported as a nonstationary upper control, not as a like-for-like competitor. A future time-indexed study must augment the state for every method rather than only for the proposed approach.

2.2 Policy-level randomness

A distribution over deterministic policies is different from a Markov stochastic policy. Sampling one deterministic policy at the start of an episode creates correlations across states and revisits. Sampling a new action from a statewise marginal on every visit removes those correlations. Both execution rules can be studied, but they must not be treated as interchangeable. Our primary rule is *sample then commit*: draw all state actions defining one deterministic policy before evaluation and preserve them throughout the episode. Per-visit marginal execution and the statewise maximum-probability policy are secondary diagnostics.

Assumption 2.2.1 (Reference measure). *The deterministic policy class has a specified probability measure μ . For a finite tabular class, μ may be uniform. For a structured or continuous class, the choice of μ is part of the model and must be reported.*

Chapter 3

Expected-Return Policy Target

For inverse temperature $\beta > 0$, define the target measure

$$P_\beta(d\pi) = \frac{\exp(\beta J_H(\pi))}{Z_\beta} \mu(d\pi), \quad Z_\beta = \int_{\Pi} \exp(\beta J_H(\pi)) \mu(d\pi). \quad (3.1)$$

Bounded rewards and a finite reference measure make Z_β finite. The parameter β carries the inverse units of reward, so a change of reward units must be accompanied by the inverse change in β . The pair (μ, β) is part of the problem specification: neither is identified by simulator data alone. For the tabular experiments, μ is uniform over complete stationary deterministic policies.

Proposition 3.0.1 (Mode preservation). *Every mode of P_β with respect to a uniform finite reference measure maximizes J_H , and every expected-return maximizer is a mode.*

Proof. The exponential function is strictly increasing, and the normalizing constant does not depend on π . □

The order of expectation and exponentiation is essential. In general,

$$\exp(\beta \mathbb{E}[G_H]) \neq \mathbb{E}[\exp(\beta G_H)]. \quad (3.2)$$

The right-hand side defines an entropic-risk quantity and places additional weight on return variability. By Jensen’s inequality it is at least the left-hand side for $\beta > 0$, with strict inequality for nonconstant return. It is not an estimator identity for the target in Equation (3.1).

The target also has the variational characterization

$$\log Z_\beta = \sup_Q \{ \beta \mathbb{E}_{\pi \sim Q}[J_H(\pi)] - \text{KL}(Q \parallel \mu) \}. \quad (3.3)$$

Thus the regularization acts on a distribution over complete policies. It is not the same as adding independent action entropy at every visited state.

Equation (3.1) may be read as a generalized posterior whose loss is $-J_H$. This interpretation is useful algorithmically, but it does not imply epistemic uncertainty calibration or Bayesian learning of the MDP. At $\beta \rightarrow \infty$, mass concentrates on expected-return maximizers; at finite β , output quality and diversity depend on both the temperature and the policy reference measure.

Chapter 4

Exact Validation Framework

Correct implementation of a gradient or MCMC transition does not establish that the intended density was specified. Validation therefore starts with finite MDPs for which all $|\mathcal{A}|^D$ stationary deterministic policies can be enumerated, where D is the number of decision states. Finite-horizon dynamic programming computes $J_H(\pi)$ for every policy, and log-sum-exp normalization gives the exact target probabilities, partition function, and action marginals in Equation (3.1).

4.1 Target-substitution diagnostic

The smallest diagnostic contains two policies with equal expected return but different return variance. Under a uniform reference measure, Equation (3.1) assigns them equal probability. A method that averages $\exp(\beta G_H)$ instead assigns more mass to the variable policy. This is a deterministic pass-or-fail

test of the order of operations rather than a performance benchmark.

Additional exact tests cover state revisitation, absorbing states, finite horizons, policy memoization, common random numbers, Metropolis–Hastings proposal ratios, replica swaps, particle ancestry, and score-function gradients. In particular, empirical frequencies of both single-chain and replica-exchange samplers must agree with the enumerated target on tiny MDPs.

4.2 Fail-fast gates

Experiments advance through checks appropriate to their inference engine. Mechanics, target definition and cost apply to every implementation; finite-bank and mixing checks apply only to the corresponding sampling variants. For variational inference, gradient correctness, optimization progress and family error replace a requirement to pass MCMC transport diagnostics:

- G1. **Mechanics.** Transition rows, rewards, termination, horizons, seeded replay, artifact writing, and budget counters pass unit tests.
- G2. **Target.** Enumeration verifies exact probabilities, modes, marginals, local and whole-policy acceptance ratios, and swap acceptance ratios.
- G3. **Approximation.** For variational inference, compare exact and sampled gradients, measure optimization progress, and characterize the restricted-family gap on enumerable instances. For fixed-bank methods, increasing the immutable tape-bank size reduces sample-average target error on repeated small problems; a failure stops simulator-only scaling.
- G4. **Mixing (MCMC only).** A one-chain rescue check first requires an

accepted, policy-changing whole-policy refresh in the upper quarter of the inverse-temperature ladder, an accepted final-edge swap, more than one cold-reaching walker, and at least two sampled cold-replica policies. A cheap two-chain ladder-flow probe then requires post-burn acceptance of at least 0.20 on every edge and an endpoint transition in each chain. Passing this probe authorizes, but does not replace, the four-chain mixing run. That full gate requires split return $\hat{R} \leq 1.05$, per-chain return ESS of at least 100, flow through every ladder edge and in both directions, between-chain coordinate agreement, and within-chain policy movement. High local acceptance or either passing fail-fast probe alone is not evidence of mixing.

G5. **Cost.** Three short runs establish memory, wall-clock time, and value-evaluation or simulator-step rates before a large job array is submitted.

The final thresholds for effective sample size and target error are frozen after pilot calibration and before confirmatory seeds are inspected. Failed, timed-out, and numerically invalid runs remain in the accounting.

Chapter 5

Expected-Return Policy Inference

5.1 Practical candidate: a distribution over complete policies

The current practical candidate is direct variational policy inference. It is distinct from the corrected MCMC and particle alternatives retained later in this chapter. The target is chosen first; an inference engine must then state its approximation and information requirements. MCMC mixing is not a prerequisite for evaluating a variational optimizer, just as a correct variational gradient is not evidence of posterior equality.

Let $\mathcal{D}$ be the decision-state set and let

$$q_\theta(\pi) = \prod_{s \in \mathcal{D}} q_{\theta,s}(\pi(s)) \quad (5.1)$$

be a categorical mean-field family. Drawing a table from this family makes its state actions independent at draw time; retaining that table makes repeated visits to the same state use the same action. Independence between coordinates therefore does not imply action resampling during execution.

The learning objective is

$$\mathcal{L}(\theta) = \beta \mathbb{E}_{\pi \sim q_\theta} [J(\pi)] - \text{KL}(q_\theta \| \mu), \quad (5.2)$$

$$\log Z_\beta - \mathcal{L}(\theta) = \text{KL}(q_\theta \| P_\beta). \quad (5.3)$$

Equation (5.3) follows by substituting the log target density and taking its expectation under q_θ . Maximizing this ELBO minimizes reverse KL within the chosen family. It does not remove the family restriction, ensure a global optimum, or make the learned distribution a calibrated posterior about unknown environment parameters.

For the uniform reference used here, the regularizer is tractable:

$$-\text{KL}(q_\theta \| \mu) = - \sum_{s \in \mathcal{D}} \sum_a q_{\theta,s}(a) \log q_{\theta,s}(a) - |\mathcal{D}| \log |\mathcal{A}|. \quad (5.4)$$

It concerns every table coordinate, including those not visited by a particular rollout. It is not the visitation-weighted entropy bonus commonly used to encourage action-level exploration.

5.2 The actual gradient estimator

Draw independent complete policies $\pi_1, \dots, \pi_N$. For each, draw K independent simulator rollouts and compute

$$\bar{G}_i = \frac{1}{K} \sum_{k=1}^K G_H(\pi_i, \xi_{ik}), \quad b_i = \frac{\beta}{N-1} \sum_{j \neq i} \bar{G}_j \quad (N > 1).$$

For a single configured policy sample the baseline is zero. Let M_{is} indicate that s appears in at least one of the K rollouts for policy i . The implementation uses

$$\hat{g} = \frac{1}{N} \sum_i (\beta \bar{G}_i - b_i) \sum_{s \in \mathcal{D}} M_{is} \nabla_\theta \log q_{\theta,s}(\pi_i(s)) - \nabla_\theta \text{KL}(q_\theta \parallel \mu). \quad (5.5)$$

The return and leave-one-out baseline are held constant when differentiating the score-function surrogate [5]. The entropy/KL contribution is differentiated exactly. Each selected policy action contributes at most one score term, irrespective of how often that state is revisited.

Proposition 5.2.1 (Unbiased masked policy gradient). *For the independent mean-field policy sampler, parameter-independent MDP dynamics, independent simulator rollouts, and the baseline above, $\mathbb{E}[\hat{g}] = \nabla_\theta \mathcal{L}(\theta)$.*

Proof. The finite policy space and bounded returns permit differentiation under the sum and rollout expectation. Without masking, the score-function identity gives the expected-return gradient. Fix a state s , condition on all other policy coordinates and simulator random numbers, and compare possible values of $\pi(s)$. Before the first visit to s , that action has not been used, so the

event of visiting s is independent of its sampled action. On the complementary event, every rollout return is independent of $\pi(s)$. Since the categorical score has zero expectation, deleting the unvisited return-score term leaves the expected gradient unchanged. The same argument gives zero mean for the masked score itself. Each b_i depends only on other independent policies and rollouts, so its product with that score has zero expectation. Finally, the exactly differentiated KL term is the required regularizer gradient. $\square$

The union mask for $K > 1$ preserves the argument: if the state is absent from all rollouts, none of their returns depends on its action. A rollout that does not visit the state may add variance when another rollout does visit it, but does not invalidate the expectation identity. The proof does not automatically extend to dependent policy draws, nonfactorized policy families, or dynamics parameterized by the same θ .

One rollout per policy is therefore sufficient for correctness. The reward signal is linear in G_H , so replacing J by an unbiased rollout preserves the gradient in expectation. This is different from replacing $e^{\beta J}$ by an exponentiated sampled return. Choosing K is a cost–variance trade-off: more rollouts per policy leave fewer independent policy draws or updates under a common interaction budget. The development comparison selects $K = 1$ for subsequent validation; it is not a universal optimum claim.

Exact tests enumerate two policy draws and all stochastic visitation outcomes on a small MDP. With $K = 1, 2$ and the baseline enabled or disabled, the expected production gradient agrees with exact ELBO differentiation to 10^{-12} . These validate the estimator, not convergence on large maps.

5.3 Initialization, training and deployment

PPO-Clip first trains a statewise guide under a separately recorded allowance. Its positive clipped probabilities initialize the variational logits. The guide is an initialization, not a replacement reference measure. Current assisted development runs use count-bonus coefficient 1.0 for guide training; refinement uses only original environment rewards. A comparison that supplies this aid only to the proposed method cannot attribute its entire performance gap to inference.

Each refinement iteration draws fresh tables and independent rollout noise, forms Equation (5.5), and updates logits with Adam. No genealogy, re-sampling weight, or sampled policy persists between updates; the learned distribution parameters and optimizer state do persist. The development settings are $N = 32$, $K = 1$, $\beta = 64$, and learning rate 0.01. Guide allocation is calibrated separately from this expected-return target.

At deployment, one table is sampled and retained for an episode. The native action-resampling marginal and deterministic marginal mode are evaluated as distinct controls. Commitment can change behavior without improving it; a single sampled wrong turn can persist on every revisit. Chapter 6 reports a large hard-map gap between these execution rules.

5.4 A strict total simulator allowance

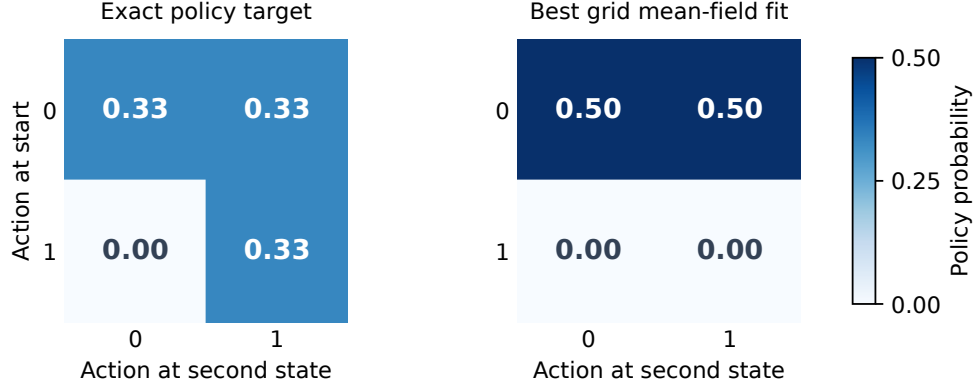
The production runner includes actual PPO-guide usage in the total cap C ; refinement receives the remainder. Before drawing a minibatch it reserves the worst-case NKH transitions. It reduces N to fit the remaining allowance or stops when fewer than two policies fit, preserving the leave-one-out baseline. A configuration explicitly requesting $N = 1$ may use one sample with a zero baseline. No long sampled trajectory is selectively discarded or truncated to satisfy the cap.

The unchanged complete-episode REINFORCE, PPO, SAC and CEM trainers can overshoot their stopping threshold by at most $H - 1$ transitions. Supplying threshold $C - H + 1$ therefore enforces the cap without changing their update equations. Double Q-learning already stops per transition. Actual usage and unused allowance are reported; a common maximum is not an assertion of identical consumption or equal wall time. Exact offline evaluation is costed separately and never feeds into training. Historical MCMC/SMC branches retain their older guide-only allowance and reject strict mode rather than silently claiming an unimplemented total cap.

5.5 Family error without poor control

Consider a two-step MDP. At the start, action 0 terminates with reward 1; action 1 visits a second state, where actions 0 and 1 terminate with rewards

High return does not certify accurate inference



Expected return is approximately 1 for both distributions.

Certified minimum family KL gap: 0.369 to 0.405 nats (beta = 64).

Figure 5.1: Exact four-policy diagnostic at $\beta = 64$. Both distributions achieve expected return essentially 1, yet the factorized family has a strictly positive certified reverse-KL gap. The displayed fit is a global grid search over Bernoulli coordinates, not a trained GridWorld policy or a claim about optimization convergence. Boundary probabilities are limits of the softmax family.

0 and 1 respectively. The four complete policies have exact return table

$$(J(a_0, a_1))_{a_0, a_1 \in \{0,1\}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

At $\beta = 64$ with a uniform reference, the exact target is essentially uniform over the three reward-one policies. A product distribution cannot put equal mass on those three policies while excluding the fourth.

A grid with 2,001 points per Bernoulli coordinate gives reverse KL 0.405465 nats and a certified global family-gap interval $[0.368818, 0.405465]$. To ob-

tain the certificate, let δ be half the grid spacing and $R = \max_{\pi} J(\pi) - \min_{\pi} J(\pi)$. Rounding each coordinate to its nearest grid point changes product-distribution total variation by at most 2δ , and changes each binary entropy by at most $h(\delta)$, where h is binary entropy in nats. Therefore the ELBO changes by at most

$$\epsilon = 2\beta R\delta + 2h(\delta).$$

Subtracting ϵ from the best grid KL gives a lower bound on the global minimum over the family, clipped at zero. The grid KL is an upper bound; continuity extends the argument to boundary limits. This diagnostic therefore exhibits family error without attributing it to an optimizer or to poor control return (Figure 5.1). It is an illustrative finite MDP, not another large-scale benchmark.

5.6 Revisits can favor either execution rule

A second exact example has one decision state and horizon 3. Action 0 terminates with reward 1; action 1 returns to the same state with reward 0.4. If action 1 has probability p , committing to a sampled table gives value $1 + 0.2p$, while resampling on each visit gives $1 + 0.4p + 0.4p^2 - 0.6p^3$. At $p = 0.75$ these values are 1.15 and 1.271875. Negating both rewards reverses the ranking. Direct checks against the exact MDP evaluators reproduce both signs. Thus consistency is a distinct decision rule, not a universal control advantage.

5.7 Alternative sampling engines

The remainder of this chapter documents target-invariant MCMC and fixed-bank SMC alternatives. Their exactness and mixing conditions are not properties of the variational optimizer above. Negative sampling results remain useful evidence about the computational structure of policy inference.

5.8 Exact and sample-average policy scores

The sampler uses a score $\tilde{J}(\pi)$. With an explicit finite transition model, $\tilde{J} = J_H$ is evaluated exactly by dynamic programming. This variant targets Equation (3.1) exactly, but is model-based.

With simulator-only access, draw an immutable bank of K random-number tapes $\Xi_K = (\xi_1, \dots, \xi_K)$ and define

$$\hat{J}_K(\pi; \Xi_K) = \frac{1}{K} \sum_{k=1}^K G_H(\pi, \xi_k). \quad (5.6)$$

Conditional on that bank, the implemented target is

$$P_{\beta, K}(d\pi \mid \Xi_K) \propto \exp\left(\beta \hat{J}_K(\pi; \Xi_K)\right) \mu(d\pi). \quad (5.7)$$

The same tape bank is reused for every policy evaluation in a chain. This is both common-random-number variance reduction and a definition of the finite-sample target. Refreshing the bank inside an ordinary Metropolis–Hastings ratio would generally invalidate detailed balance. For finite policy spaces and

bounded returns, $\widehat{J}_K$ converges uniformly to J , so Equation (5.7) converges to Equation (3.1); finite K is nevertheless an approximation, not an unbiased exact-target construction.

5.9 A learned proposal that cannot change the target

PPO-Clip [6] is trained first and then frozen. Its statewise action probabilities guide local moves, but PPO is neither the target nor an uncorrected component of the final sampler. Let $p_\phi(a \mid s)$ be the frozen probabilities and define

$$g_{\eta,\phi}(a \mid s) = \eta p_\phi(a \mid s) + (1 - \eta) \frac{1}{|\mathcal{A}|}, \quad 0 \leq \eta < 1. \quad (5.8)$$

The uniform component gives every action positive probability. If the current action is a , a proposed different action a' has probability

$$q_s(a' \mid a) = \frac{g_{\eta,\phi}(a' \mid s)}{1 - g_{\eta,\phi}(a \mid s)}, \quad a' \neq a. \quad (5.9)$$

Both the guide snapshot and η are stored with the run. Setting $\eta = 0$ gives the uniform-proposal ablation.

5.10 Single-site policy Metropolis–Hastings

One local proposal selects a decision state uniformly, changes only its action according to Equation (5.9), and leaves all other policy entries fixed. At

inverse temperature b , accept π' from π with probability

$$\alpha_b(\pi, \pi') = \min \left\{ 1, \exp \left[b \{ \tilde{J}(\pi') - \tilde{J}(\pi) \} \right] \frac{\mu(\pi') q_s(a \mid a')}{\mu(\pi) q_s(a' \mid a)} \right\}. \quad (5.10)$$

The state-selection terms cancel. The experiments use uniform μ , so its ratio is one. The remaining Hastings ratio is essential because forcing a different action makes even a fixed categorical guide asymmetric. With full support, this kernel is irreducible on the finite policy space and satisfies detailed balance with the chosen exact or fixed-tape target.

5.11 Exact whole-policy independence refresh

Single-site edits can be ineffective on a sparse-return plateau, where many policies have the same score but reaching a useful policy requires several coordinated action changes. Each active replica update therefore chooses the single-site kernel with probability $1 - \rho$, or an independence proposal with probability ρ . The fixed whole-policy proposal is

$$Q_\phi(\pi) = w_{\text{map}} \mathbf{1}\{\pi = \pi_\phi^{\text{MAP}}\} + w_{\text{guide}} \prod_{s \in \mathcal{D}} p_\phi(\pi(s) \mid s) + w_{\text{uniform}} |\mathcal{A}|^{-|\mathcal{D}|}, \quad (5.11)$$

where $\mathcal{D}$ is the decision-state set and π_ϕ^{MAP} is the statewise mode of the frozen PPO guide. The defaults are $\rho = 0.10$ and $(w_{\text{map}}, w_{\text{guide}}, w_{\text{uniform}}) = (0.25, 0.70, 0.05)$. The components overlap, so the proposal density is the sum in Equation (5.11), evaluated by log-sum-exp, not merely the density of the component that generated the draw. The positive uniform component

guarantees full support.

At inverse temperature b , an independent draw $\pi' \sim Q_\phi$ is accepted with

$$\begin{aligned} \log R_b(\pi, \pi') &= b\{\tilde{J}(\pi') - \tilde{J}(\pi)\} \\ &\quad + \log \mu(\pi') - \log \mu(\pi) \\ &\quad + \log Q_\phi(\pi) - \log Q_\phi(\pi'), \end{aligned} \tag{5.12}$$

$$\alpha_b^{\text{global}}(\pi, \pi') = \min\{1, \exp[\log R_b(\pi, \pi')]\}. \tag{5.13}$$

This is an ordinary independence Metropolis–Hastings step [7]. The complete proposal ratio is mandatory. Consequently, the guide, its mode, and the branch probability affect efficiency but cannot change the invariant target. Full support gives an asymptotic finite-space convergence guarantee, not evidence that a finite run has mixed.

Forced-different proposals can be periodic in a two-action domain. We therefore mix every complete within-replica-plus-swap sweep with the identity kernel:

$$K_\delta = \delta I + (1 - \delta)K, \quad 0 < \delta < 1. \tag{5.14}$$

The stay-put decision is independent of the chain state, so it preserves the product target while giving every augmented chain state a positive self-transition. The default is $\delta = 0.05$; swap parity advances only on active sweeps. The branch settings, local and global proposal/acceptance counts, a separate count of accepted policy-changing global moves, δ , and the realized lazy-sweep count are stored with each run.

5.12 Fixed structural blocks and connectivity

The implementation also supports two separately selected branches: a path-only update and an ordered sweep over a fixed catalog of blocks. The catalog follows the modal guide from the start to a terminal state and then partitions its one-hop recovery neighbors into pairs (with a possible final singleton). This construction queries exact successor probabilities and is therefore model-assisted even when the scores use fixed tapes. A cycling modal path is rejected, rather than silently selecting a different construction from production samples.

The path block uses a MAP/guide/uniform independence mixture restricted to its coordinates. Its Hastings ratio sums all overlapping mixture components. For a small recovery block, let m denote the guide-MAP assignment and let R be the guide/uniform mixture conditioned on not choosing m . The proposal moves from any $x \neq m$ to m deterministically, and from m to $y \sim R$. Its reverse-to-forward proposal ratio is $R(x)$ in the first case and $1/R(y)$ in the second. Applying the corresponding Metropolis–Hastings correction preserves the conditional policy target. Composing the fixed block kernels preserves invariance, although the complete ordered sweep need not be reversible.

Invariance alone does not establish connectivity. If local and whole-policy branches are disabled, the active catalog must cover every decision coordinate. Even full coverage can leave a deterministic ordered sweep trapped in a parity class on a flat two-action problem. To prevent this, each scored local or structural substep uses

$$\alpha_{\text{substep}}(x, y) = 0.95 \min\{1, \exp[L(x, y)]\}.$$

This is a mixture of the ordinary MH kernel with an independent identity step. It preserves detailed balance and permits individual replicas and blocks to remain fixed while others move. Whole-sweep laziness alone cannot provide this property. Rejected substeps still count their proposal and score evaluation, so the acceptance accounting remains explicit.

The structural kernels have passed finite-policy implementation checks, but their medium-scale development runs have not established useful mixing. They are recorded experimental variants, not a validated scalability result.

5.13 Replica exchange across inverse temperatures

Single-site moves can mix slowly when good complete policies occupy separated modes. Replica exchange, also called parallel tempering, couples chains at inverse temperatures

$$0 = b_0 < b_1 < \cdots < b_{M-1} = \beta, \quad b_\ell^{(0)} = \beta \{\ell / (M - 1)\}^p, \quad (5.15)$$

where $p > 0$ controls the initial ladder spacing. Development-only pilots may replace this power ladder with a fixed, explicitly listed ladder that places more replicas at measured exchange bottlenecks, following the feedback principle of Katzgraber et al. [8]. Adaptation stops before target samples are collected, and the endpoints remain zero and β . Replica ℓ applies the fixed mixture of Equations (5.10) and (5.13) with $b = b_\ell$. At a declared interval, four

checkerboard swap sub-sweeps are applied by default, with parity alternating globally. These extra swaps reuse cached values and compose invariant kernels, so they can improve temperature-space flow without extra policy evaluations [9]. For replicas $i < j$, the swap acceptance probability is

$$\alpha_{i,j}^{\text{swap}} = \min\left\{1, \exp\left[(b_j - b_i)\{\tilde{J}(\pi_i) - \tilde{J}(\pi_j)\}\right]\right\}. \quad (5.16)$$

This is a Metropolis update for the product of tempered targets

$$\prod_{\ell=0}^{M-1} \mu(\pi_\ell) \exp(b_\ell \tilde{J}(\pi_\ell)),$$

and hence preserves the desired cold marginal at $b_{M-1} = \beta$ [10]. The PPO guide affects only proposal efficiency; Equations (5.10), (5.13), and (5.16) retain the target.

By default, all replicas start at the PPO statewise mode, while the $b_0 = 0$ chain is allowed to forget that initializer during burn-in. A separate overdispersed diagnostic initializes hotter replicas from the policy prior. Initialization cannot alter the stationary distribution, but it can strongly affect a finite run. Only thinned post-burn-in draws from the b_{M-1} replica are reported as target samples.

5.14 Diagnostics and cost accounting

Each run records the beta ladder, local and global proposals and acceptances, accepted policy-changing global moves, and adjacent swap proposals and

acceptances for both the complete and post-burn windows, retained returns, return effective sample size, policy-value evaluations, lazy sweeps, simulator transitions, PPO training transitions, wall-clock time, and software and source hashes. Return effective sample size is necessary but can miss movement between policies with similar value. We therefore also record conservative and quantile summaries of action-indicator effective sample sizes, statewise switch-rate summaries, distinct sampled policies, walker temperature visits, endpoint transitions, round trips, and the number of walkers reaching the cold replica. Independent warm and overdispersed chains provide the between-chain check. Calibration uses two distinct decisions. A short two-chain probe tests only whether the proposed ladder has viable post-burn edge exchange and endpoint flow; passing it merely permits the longer run. The full multichain gate then tests return $\widehat{R}$ and ESS, edge exchange, cold-reaching walkers and round trips, coordinate total variation, unique-policy fraction, mean policy Hamming movement, and switching at nondegenerate coordinates. Only the latter supports a mixing claim. A high-return cold replica cannot compensate for a failure at either stage.

For M replicas and scored proposal counts N_{local} , N_{global} , N_{path} , and N_{block} , summed over every replica and block repetition, logical policy-value evaluations are

$$M + N_{\text{local}} + N_{\text{global}} + N_{\text{path}} + N_{\text{block}}.$$

An ordered structural sweep can perform multiple evaluations per replica. Exact repeated-policy cache hits and actual score computations are logged separately and sum to this logical total. Swaps reuse current scores.

PPO training is part of the proposed method’s cost. Matched-budget comparisons therefore include every interaction and optimizer update used to train the guide plus all downstream policy evaluations. Exact dynamic-programming evaluations are logged separately because they consume model computations but zero simulator transitions.

5.15 Complementary estimators

Direct variational inference is defined at the start of this chapter in Equations (5.2)–(5.5). As a separate alternative, replicated policy SMC instantiates each state action once in a lazy policy memo and advances K simulator replicas under that same policy. For newly exposed states U_t , its incremental log weight is

$$\Delta \log w_t = \frac{\beta}{K} \sum_{k=1}^K r_{t,k} + \sum_{s \in U_t} [\log \mu(a_s \mid s, m_{<s}) - \log q(a_s \mid s, m_{<s})]. \quad (5.17)$$

The mean reward is formed before exponentiation, proposal correction is applied once per newly assigned policy entry, revisits reuse the memo, and resampling copies the entire ancestry-dependent state. Sequential Monte Carlo can define powerful variational families [11] and particle objectives [12], but those results do not by themselves make an exponentiated noisy return an unbiased estimator of Equation (3.1). Here replicated SMC is evaluated against the explicitly stated fixed-tape target and against exact enumeration. Diagnostic controls include unguided and PPO-guided MCMC/SMC and PPO alone. All simulator interactions are counted.

Chapter 6

Experimental Design

The empirical study separates four questions: (i) does the implementation estimate the stated objective correctly, (ii) what approximation or optimization error remains, (iii) how well does the learned distribution control the environment under each execution rule, and (iv) what simulator and compute cost is required? Direct variational learning is the practical candidate. Sampling alternatives additionally require target-fidelity and mixing diagnostics; their failures are not silently reinterpreted as successful inference.

6.1 Domains and scope

Stochastic GridWorld is the planned primary study. Easy, medium, and hard families use, respectively, 8×8 , 16×16 , and 32×32 grids; wall fractions of 9.4%, 19.9%, and 30.0%; action success probabilities of 0.90, 0.80, and 0.75; and horizons 64, 160, and 384. Frozen-map audits record reachability, shortest-

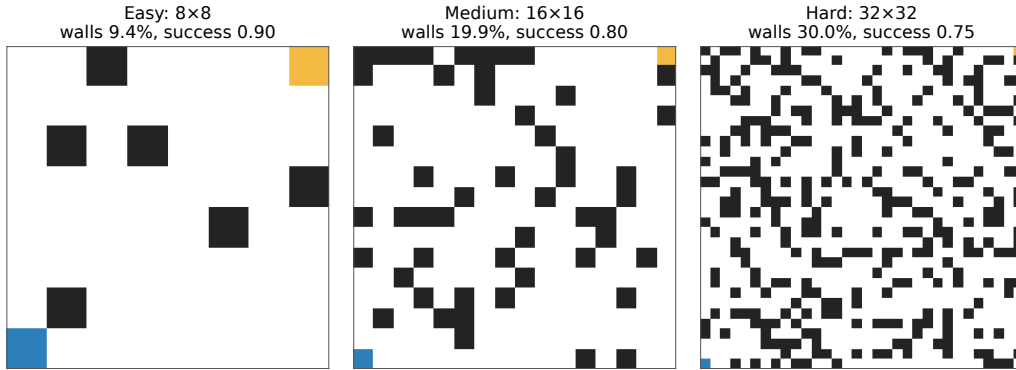


Figure 6.1: Representative development maps for the three GridWorld tiers.

White cells are traversable, dark cells are walls, blue marks the start, and gold marks the goal. Held-out maps are frozen separately and remain unused during method development and tuning.

path length and count, and single-cell start–goal bottlenecks. Seed zero maps are development-only; held-out maps are frozen before confirmation. Small instances retain exact enumeration; larger instances retain exact dynamic-programming evaluation of any fixed policy even when enumerating all policies is impossible.

Only domains that add a distinct scientific stress test are considered after GridWorld passes its medium gates:

A Tireworld extension would use a verified factored-state adapter; pyRDDL-Gym is a possible RDDL-to-simulator route [16]. Transition, termination, reward, and action-mask tests on a small instance are mandatory before any such results are included.

Table 6.1: Domain-extension decision. Extensions are conditional on the primary GridWorld study rather than required to enlarge the result table.

Domain	Added evidence	Decision
Blackjack	Compact nonspatial stochasticity and intrinsic return variance with two actions [13]	First low-cost extension after GridWorld
Tireworld	Stochastic failures, resources, sparse success, and competing routes [14]	Best second headline domain if time and compute remain
Taxi	Action masks and subgoals, but mostly deterministic and spatial	Optional smoke test only
Academic Advising	Factored multivalued state and concurrent actions [15]	Future structured-policy work
Continuous control	Requires a new reference measure and policy-space sampler	Out of scope

6.2 Competitors and controls

The competitor implementations preserve each published core update while using a common tabular code path for identical environment semantics, seeding, transition accounting, and artifacts. They are named by the exact implemented variant rather than by a broader software package.

The exact control is nonstationary and may use the known model, so it is an upper reference rather than a fair learned competitor. All learned and inferred output policies use stationary physical-state inputs. For correct finite-horizon backup targets, discrete SAC uses time-indexed twin critics and replay tuples, while Double Q-learning uses time-indexed tables and projects its result into the declared stationary policy class using state-time visitation.

Comparisons are stratified by information access. In the known-model stratum, exact-value tempering is tested for target fidelity and mixing against

Table 6.2: Learned competitors. Uniform random behavior and exact finite-horizon dynamic programming are controls, not competitors.

Reported method	Implemented variant	Role
REINFORCE	Monte Carlo policy gradient with learned state-value baseline [5]	On-policy score-function baseline
PPO-Clip	Clipped surrogate, Monte Carlo advantages, entropy bonus; no GAE or value clipping [6]	On-policy clipped-surrogate baseline and separately costed proposal trainer
Discrete SAC	Categorical actor, twin critics, target critics, replay, and fixed entropy temperature [17]	Off-policy entropy-regularized baseline
Categorical CEM	Distribution search over complete stationary deterministic policies [18]	Direct black-box policy search
Double Q-learning	Online tabular Double Q-learning with linearly annealed epsilon-greedy exploration [19]	Value-learning baseline

exact enumeration or known-model controls. Fixed-tape scoring alone does not imply simulator-only access: the current path/recovery block builder reads exact successor probabilities. Such runs remain model-assisted even when scores come from rollouts. In a simulator-only stratum, that builder must be disabled or replaced by separately counted simulator-based construction. Eligible fixed-tape variants, direct variational learning with its PPO initializer, and the five learned competitors receive generative-simulator access. Exact-model tempering may be displayed beside simulator-only learners as a descriptive reference, but such a cross-stratum table cannot support a sample-efficiency superiority claim. Model value evaluations, simulator transitions, and guide training transitions are therefore reported in separate cost columns.

6.3 Common evaluation semantics

For the committed-policy study, the current evaluator samples a complete deterministic policy from the reported distribution and commits to it for the entire episode. For an explicit particle or MCMC sample, that complete policy is used directly. For statewise categorical outputs, one action is sampled once at every decision state before evaluation. Primary metrics are mean exact expected return and exact goal-reaching probability over these committed policies. For independent VI or baseline policy draws, the sample standard deviation describes policy variation and the standard error describes conditional evaluation precision. Neither measures uncertainty across training runs. Correlated MCMC draws or resampled particles require separate dependence and ancestry diagnostics; the independent-draw formula is not valid for them.

This rule is an intervention on stochastic baseline outputs, not their native deployment semantics. Native per-visit stochastic value is therefore a co-primary result for REINFORCE, PPO, and SAC in practical control comparisons; their committed-policy values answer a narrower policy-class question. A sampler’s statewise marginals cannot replace its committed-policy result because they remove policy-level dependence. The statewise maximum-probability policy is also reported separately. Secondary metrics include lower-tail return, distributional target error where enumeration is possible, and mode recovery.

6.4 Fair budgets and tuning

Every comparison uses the same maps, training seeds, evaluation streams, horizon, reward, and termination convention. Current GridWorld runs use $\gamma = 1$; discounted baseline semantics must be validated before any $\gamma < 1$ experiment. Method-specific tuning grids are allowed because a single global learning rate is not a fair tuning policy, but tuning allowance is matched in environment transitions and core-hours. Tuning maps and seeds are disjoint from the final confirmatory set.

The inverse temperature defines the target and is not selected after observing held-out control performance. Exact diagnostic maps use a predeclared sensitivity grid; the headline GridWorld target freezes one value on seed-zero development maps. The current development value is $\beta = 64$. Any additional value is reported as target sensitivity with appropriate approximation and cost diagnostics, not as a best-of-grid result.

All interactions used by PPO guide training, variational refinement, SMC, policy-bank evaluation, and sample-average MCMC count toward the proposed method. Exact policy-value evaluations, optimizer updates, wall time, CPU or accelerator time, and peak memory are reported separately. A generally applicable exploration aid is offered to compatible methods or labeled as a separate assisted ablation. Competitor bugs are fixed; no baseline is weakened to improve a comparison.

New variational and baseline runs enable `strict_transition_budget`: the declared allowance is a maximum for all training transitions, including the initial-

izer. Refinement receives the total minus actual guide usage. Complete-return reservation can leave a small unused allowance, which is reported. Historical tempering configurations retain different semantics: their `transition_budget` covers guide training, and subsequent sampling adds cost. They are not relabeled as strict-cap comparisons. Full-budget PPO under the same allowance is an essential attribution control; the current control starts from scratch, rather than resuming the guide optimizer.

6.5 Cost-effective execution order

After the claim-specific checks in Section 4, one seed and the smallest budget run for every method on each map family. Only finite, semantically valid runs advance to a three-seed timing and learning pilot. That pilot estimates resource requests and identifies clearly dominated settings. Hyperparameters are then tuned on designated development instances. Finally, one frozen manifest launches the full seed-by-map comparison. Job arrays are capped to avoid simultaneous infrastructure failures, and each completed run writes a validated marker so interrupted arrays can resume without overwriting results.

The same order applies to domain extension: one exact Blackjack instance, one seed per method, then three-seed timing, and only then a complete table. Tireworld work begins only after the GridWorld confirmatory manifest is frozen.

Table 6.3: Medium GridWorld development comparison: one map, training seed 572, and a 1.2M total-transition cap. Parentheses are Monte Carlo standard errors over 256 committed-policy draws, not uncertainty across training runs. Native means action resampling. Wall time includes offline evaluation.

Method	Committed (SE)	Native	Seconds
VI (PPO-initialized)	0.725 (0.044)	0.992	40.9
PPO + count bonus	0.656 (0.098)	1.015	109.9
PPO	0.008 (0.137)	0.766	103.9

6.6 Current empirical status

The confirmatory study remains incomplete. All numerical results below are development diagnostics; held-out maps have not supplied tuning decisions.

6.6.1 Variational allocation and full-budget controls

A cached medium-map PPO guide trained for 200,015 transitions initialized three independent refinements per rollout setting. With approximately one million additional transitions, $K = 1$ produced committed returns 0.581, 0.595, and 0.548; $K = 4$ produced -0.144, -0.244, and -0.087. The corresponding update counts were 662–669 versus 157–160. Every refinement improved its paired guide estimate and estimated ELBO. Evaluation used 512 independent policy draws, coupled within each guide/refinement pair. These refinements share one guide and map, so they do not establish robustness across independently trained guides.

Fresh-guide, strict-cap runs then used nominally 200k guide transitions and 1M refinement transitions. On the easy map, committed value rose from the

guide’s 3.010 to 3.318, below the nonstationary upper control 3.451. Actual usage was 1,199,880 transitions. The medium run used 1,199,725 transitions and achieved committed value 0.725 versus its guide’s -0.225. Table 6.3 includes full-budget PPO controls at the same medium training seed 572. The VI guide used count-bonus coefficient 1.0; refinement used original rewards. Assisted PPO used coefficient 1.0 throughout, and standard PPO used zero. Both controls trained from scratch.

VI’s committed value is slightly above assisted PPO’s and its native value is slightly below. The committed difference 0.069 is smaller than the combined conditional evaluation standard error. Neither equivalence nor superiority is established. The shorter local runtime is likewise one observation, not a general speedup claim. These runs share executable snapshot and map; their wall times include offline exact evaluation and use one numerical-library thread.

6.6.2 Hard-map exploration and execution semantics

Figure 6.2 displays the same completed development runs under both execution rules. The hard-panel allocation change is separately labeled; it is not an increase in the total simulator allowance.

The unchanged 200k-guide allocation failed on hard map zero with seed 573: all guide and refinement batches returned the no-goal value -38.4. The estimated ELBO rose through reduced KL, without improved control. The map is not unreachable: its finite-horizon upper return control is -4.001.

A separately frozen two-step diagnostic first trained full-budget assisted PPO. Its first successful batch occurred near 538k transitions, beyond the original guide allowance. Native goal probability reached 0.995. This satisfied the predeclared condition for testing a 1M-guide/200k-refinement reallocation at the same total cap, map, and seed. That run consumed 999,721 guide and 199,549 refinement transitions. It achieved committed return -19.219 (conditional MCSE 0.679), compared with its guide’s -23.077. Native return was -6.775.

Committed goal probability reached 0.700, versus zero with the short guide; native action resampling reached 0.993. Full-budget assisted PPO achieved committed return -20.900 and native return -6.415, with goal probabilities 0.637 and 0.995 respectively. Wall times were 132.4 seconds for reallocated VI and 133.9 seconds for assisted PPO, including evaluation. This single-seed allocation diagnostic supports testing independent guide seeds, not a general hard-grid success or superiority claim. The large execution gap is also a counterexample to treating policy commitment itself as a control advantage.

6.6.3 Sampling alternatives: development findings

Preliminary medium-GridWorld exact-value tempering diagnostics produced high-return committed policies but exposed a sharp intermediate-temperature exchange bottleneck. Four independent 32-temperature, power-four chains, including an overdispersed chain, each retained 800 draws. They failed the predeclared gate: split return $\hat{R} = 1.096$, minimum per-chain return ESS of 10.5, minimum whole-run adjacent-swap acceptance of 0.049, no completed

round trip, and maximum pairwise state-action total variation of 0.699. The principal warm-chain bottleneck lay between inverse temperatures 1.44 and 1.98. Because the legacy acceptance counter included burn-in, subsequent runs also store a separate post-burn rate; the ESS, round-trip, and between-chain failures do not depend on that counter. A subsequent two-chain, 4,000-iteration custom-ladder probe also failed. Its warm chain had post-burn acceptance between 0.598 and 0.997 on every edge, but no walker traversed both endpoints. In the overdispersed chain, replicas below the cold endpoint remained on a flat random-policy plateau near return -16 , while the cold replica remained near return 1.1; the final edge consequently accepted no swap. High acceptance inside the bad plateau therefore did not indicate useful mixing.

Subsequent whole-policy and structural-block rescue attempts did not establish adequate mixing. An isolated path-proposal screen used four independent 4,500-step chains at $\beta = 3.42$, discarding 1,200 steps and retaining every tenth subsequent state. Low-initialized chains remained near return -16 ; high-initialized chain medians were approximately -1.56 and -1.41 . Path banks fitted on one chain from each region failed to generate useful bidirectional crossings on the other chains. These measurements do not estimate the relative posterior mass of the regions.

A bounded three-seed guide-field replica-exchange screen subsequently found high-return policies but did not validate sampling adequacy. Return effective sample sizes were 28–66, split $\hat{R}$ was 1.032, maximum between-chain coordinate total variation was 0.578, and completed round trips were 0, 1, and

Table 6.4: Required confirmatory reporting. No entry is populated from pilot runs.

Question	Comparison	Primary report
Estimator correctness	exact differentiation	gradient error under stochastic visitation
Family error	enumerable posterior	variational gap and omitted dependence
Sampling alternatives	chains and ladders	ESS, swaps, ancestry, and agreement
Control quality	five competitors	native and committed return and goal probability
Robustness	map families and seeds	uncertainty and failed-run count
Scalability	matched accounting	interactions, time, and memory

1. A fixed-bank SMC screen retained only 3, 4, and 4 root ancestors from 256 particles and returned values near -16. These alternatives have different costs and information access; they are not pooled into a simulator-efficiency leaderboard. Historical failed checks remain failed. The practical VI candidate instead requires family-error, learning-stability, and execution-aware control checks before full-scale confirmation.

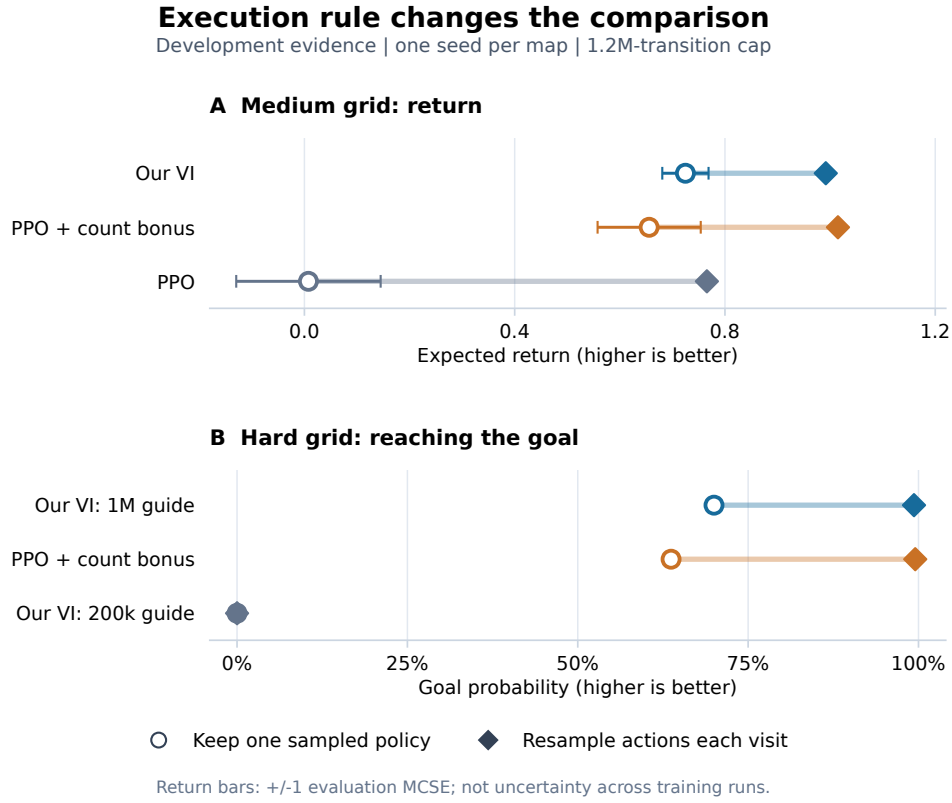


Figure 6.2: Execution-aware development comparison with a common 1.2M-transition cap. The medium panel uses seed 572 and the hard panel seed 573, each on map zero. Open circles commit to one sampled policy; filled diamonds resample actions on each visit. The return error bars are one conditional evaluation standard error over 256 policy draws, not training-level uncertainty. The hard-panel guide allocations retain the same total allowance. PPO with the count bonus is an explicitly assisted control; standard PPO is separate. These single-seed results motivate, but do not replace, multi-seed comparison.

Chapter 7

Discussion and Limitations

7.1 Relation to policy-gradient learning

The current practical method uses a score-function policy gradient, but its random object is a complete deterministic policy table and its regularizer acts on a distribution over those tables. The PPO guide initializes the variational distribution once. Subsequent updates change its logits directly using original rewards; fresh policies are sampled each iteration. Neither trajectory weights nor particle genealogy are carried across updates. The relation to REINFORCE is therefore explicit, not denied: the difference lies in the sampled object, objective, and execution semantics.

The separately implemented MCMC alternative behaves differently. Its guide is frozen while complete policies evolve through corrected proposals and replica swaps; there is no guide-gradient update during that chain. Consequently, claims about invariant distributions or MCMC mixing must not be used as

guarantees for the variational optimizer.

Determinism is a representation and execution choice, not a general computational advantage. It makes repeated actions consistent but does not remove stochastic transitions or the exponentially large policy space. On the hard development instance, action resampling gives much higher goal probability than commitment for both VI and PPO. A committed wrong action can persist at every revisit. This is a substantive behavioral difference and a limitation for control, not a result to hide by reporting only the more favorable execution rule.

7.2 Approximation and interpretation

With a uniform reference, modes of the target maximize expected return. A finite-temperature draw need not do so. The resulting dispersion depends on the reference and temperature; it is not calibrated epistemic uncertainty about an unknown environment.

The direct ELBO avoids exponentiating noisy return estimates, but its gradient can have high variance. Mean-field policies cannot express general cross-state dependence, and reverse-KL optimization may represent one policy region while omitting others. An unbiased estimator is a statement about the expected update at fixed parameters, not a guarantee of global optimization or accurate posterior coverage. Family error needs its own small-instance characterization and behavioral tests.

Replicated SMC retains policy ancestry under an explicitly fixed-bank approx-

imation but can degenerate severely. The tested medium configuration kept only a few root ancestors and almost never reached the goal. Completing unvisited table entries makes policies syntactically distinct without establishing useful behavioral diversity.

For MCMC, exact Hastings correction guarantees invariance, not finite-run mixing. Local edits and even global proposals can fail to traverse high-dimensional policy modes. Adjacent swap acceptance, return ESS, coordinate movement and agreement between dispersed chains must therefore be checked together. The tested bridge produced good policies but failed to establish full distributional agreement.

7.3 Exploration and guide attribution

Sparse rewards create a common exploration problem [20]. In the hard development run, a 200k-transition guide never achieved success, and all refinement returns were equal. The entropy term then improved the estimated ELBO without improving control. Full-budget assisted PPO found success only after that initial allowance. Reallocating the same total budget toward guide learning helped in a subsequent conditional development test. Repetition across independent guides is required before treating this allocation as a robust choice.

This dependence on a guide is part of the method’s cost and limitation. Improvement over an under-budget initializer does not establish improvement over full-budget PPO. Standard PPO and exploration-assisted PPO

are separate controls. In the assisted comparison, PPO retains the count bonus throughout training, whereas VI uses it only during initialization; this distinction must remain visible. Performance against an unassisted baseline alone cannot isolate an inference contribution.

7.4 Access, policy class and scope

Direct variational refinement uses only simulator returns for learning. Exact dynamic programming is used offline for evaluation. In contrast, exact-score MCMC uses model information during inference, and the optional structural block builder reads exact successor probabilities even when its scores use rollout tapes. These are separate information-access strata, not interchangeable sample-efficiency competitors.

The fixed-bank target converges to the intended expected-return target under finite-policy bounded-return assumptions, but finite bank size remains an approximation. Bounded returns also permit a nonnegative Poisson estimator for a pseudo-marginal construction; Appendix A gives its identity and variance. Its potentially prohibitive cost prevents mathematical exactness from serving as a practicality claim.

All reported output policies are stationary in physical state, while the finite-horizon dynamic-programming upper control is nonstationary. PPO uses Monte Carlo advantages without GAE or value clipping. Discrete SAC uses a fixed entropy temperature and time-indexed twin critics; Double Q-learning uses time-indexed tables before an occupancy-weighted projection to

a stationary output. These precisely specified implementations preserve their core updates but do not claim every enhancement of the algorithm families. Current GridWorld studies use $\gamma = 1$.

Finally, the present development samples cannot establish generality or practical equivalence. Independent training seeds and held-out maps are the experimental units; repeated policy draws only improve conditional evaluation precision. A nonspatial Blackjack study could add evidence at low cost, whereas Tireworld would test structured stochastic failure with greater implementation effort. These extensions cannot replace completing the primary comparison and its uncertainty analysis.

Chapter 8

Conclusion

Expected-return policy inference is meaningful only after the policy space, reference measure, inverse temperature, and execution rule are explicit. Averaging environment randomness before exponentiation distinguishes its target from an exponential-trajectory objective. This established Gibbs construction motivates an implementation whose approximations can be tested independently of its control performance.

The practical candidate uses simulator-based variational learning over complete policy tables, initialized by a separately costed PPO guide. Its masked score-function estimator is unbiased under the stated sampling assumptions, and exact small-instance tests exercise the actual implementation. It does not follow that a finite run globally optimizes the objective or represents the full target. Corrected MCMC and fixed-bank SMC remain documented alternatives with separate invariance, approximation and mixing requirements. Development results identify useful questions rather than settle the final

comparison. One rollout per policy permits more effective refinement than four in the tested allocation. A medium-map matched-budget check gives similar values to assisted PPO, while a hard-map failure and conditional budget reallocation expose the importance of exploration before refinement. The sharp difference between commitment and action resampling also prevents equating policy consistency with improved control.

Strict total simulator caps and explicit execution semantics now support a reproducible confirmatory study. The remaining evidence must characterize family error, independent-guide variability, held-out performance and full cost against the five declared baseline families. Competitive or mixed results can support a useful policy-inference contribution; neither universal superiority nor calibrated posterior uncertainty is claimed.

Appendix A

Core Derivations

Gibbs variational identity. Let P_β be the target in Equation (3.1). For any $Q \ll \mu$,

$$\text{KL}(Q \| P_\beta) = \mathbb{E}_Q \left[\log \frac{dQ}{d\mu} - \beta J_H(\pi) + \log Z_\beta \right] \quad (\text{A.1})$$

$$= \log Z_\beta - (\beta \mathbb{E}_Q[J_H(\pi)] - \text{KL}(Q \| \mu)). \quad (\text{A.2})$$

Non-negativity yields Equation (3.3), with equality at $Q = P_\beta$.

Why a rollout is valid in the direct objective. For $\pi \sim q_\theta$ and $\tau \sim p(\cdot | \pi)$, iterated expectation gives

$$\mathbb{E}_{\pi, \tau}[G_H(\tau)] = \mathbb{E}_{\pi \sim q_\theta}[J_H(\pi)].$$

No exponential or logarithm is applied between the inner and outer expectations. Applying the score-function identity to Equation (5.2), and dropping

terms whose expected score is zero, gives Equation (5.5).

Local detailed balance. At inverse temperature b , write

$$\tilde{P}_b(\pi) \propto \mu(\pi) \exp(b\tilde{J}(\pi)).$$

The Metropolis–Hastings ratio is

$$\frac{\tilde{P}_b(\pi')q(\pi \mid \pi')}{\tilde{P}_b(\pi)q(\pi' \mid \pi)} = \exp\left[b\{\tilde{J}(\pi') - \tilde{J}(\pi)\}\right] \frac{\mu(\pi')}{\mu(\pi)} \frac{q_s(a \mid a')}{q_s(a' \mid a)}.$$

The uniform decision-state selection cancels, producing Equation (5.10). Full support makes every one-coordinate change possible, and a finite sequence of such changes connects any two tabular policies.

Replica-swap ratio. Before a swap, the factors for replicas i and j are proportional to $\mu(\pi_i) \exp(b_i \tilde{J}(\pi_i)) \mu(\pi_j) \exp(b_j \tilde{J}(\pi_j))$. After exchanging the two policies, the reference-measure factors cancel and the new-to-old density ratio is

$$\exp\left[(b_j - b_i)\{\tilde{J}(\pi_i) - \tilde{J}(\pi_j)\}\right],$$

which gives Equation (5.16). Alternating disjoint adjacent pairs changes scheduling efficiency but not this invariant product distribution.

Fixed-tape convergence. For every fixed policy, the strong law gives $\hat{J}_K(\pi) \rightarrow J(\pi)$ almost surely. Because the tabular policy space is finite, convergence holds simultaneously for all policies. Continuity of exponentiation

and finite normalization then imply $P_{\beta,K}(\pi \mid \Xi_K) \rightarrow P_\beta(\pi)$ for every policy. This argument justifies a convergence study in K ; it does not make a finite tape bank exact for P_β .

A nonnegative unbiased policy weight from bounded rollouts. Suppose a simulator return satisfies $G_\pi \geq a$ almost surely, where the finite-horizon reward specification supplies the known constant a . For $\lambda > 0$, draw $N \sim \text{Poisson}(\lambda)$ and, conditional on N , independent returns $G_{\pi,1}, \dots, G_{\pi,N}$. Define

$$\widehat{W}_\lambda(\pi) = \exp(\beta a) \prod_{i=1}^N \left(1 + \frac{\beta(G_{\pi,i} - a)}{\lambda} \right). \quad (\text{A.3})$$

Every factor is nonnegative. Conditioning first on N , then using the probability generating function of a Poisson variable, gives

$$\mathbb{E}[\widehat{W}_\lambda(\pi)] = \exp(\beta a) \mathbb{E}_N \left[\left(1 + \frac{\beta(J(\pi) - a)}{\lambda} \right)^N \right] \quad (\text{A.4})$$

$$= \exp(\beta a) \exp(\beta(J(\pi) - a)) = \exp(\beta J(\pi)). \quad (\text{A.5})$$

Thus a pseudo-marginal Metropolis–Hastings chain on the policy and estimator randomness has the exact policy target as its marginal [21, 22]. The identity is not automatically an efficient algorithm. Its expected rollout count is λ , and direct calculation gives

$$\frac{\mathbb{E}[\widehat{W}_\lambda(\pi)^2]}{\exp(2\beta J(\pi))} - 1 = \exp\left(\frac{\beta^2 \mathbb{E}[(G_\pi - a)^2]}{\lambda}\right) - 1. \quad (\text{A.6})$$

Bounded relative variance requires rollout cost scaling with $\beta^2 \mathbb{E}[(G_\pi - a)^2]$, potentially prohibitive.

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בחירת מדיניות באמצעות הסקה לפי תוחלת התגמול

Eden Havri

עבודת גמר לתואר מוסמך למדעי הטבע

אוניברסיטת בן-גוריון בנגב

2026 (research draft)

תקציר

אנו חוקרים הסקה בייסיאנית מוכללת מעל מדיניות דטרמיניסטיות מלאות בתהליך החלטה מרקובי סטוכסטי בעל אופק סופי. התפלגות המטרה נקבעת לפי תוחלת התגמול: תחילה מחשבים תוחלת ביחס לאקראיות בסביבה, ואז מפעילים פונקציה מעריכית. הגדרה זו, המבוססת על עבודות קודמות, שונה ממתן משקל מעריכי לתגמול שהתקבל במסלול אקראי יחיד.

השיטה המעשית הנבחנת לומדת התפלגות מפורקת מעל טבלאות מדיניות מלאות. אלגוריתם PPO מספק אתחול בעל עלות מדווחת בנפרד. לאחריו מתבצע עדכון וריאציוני ישיר בעזרת תגמולים מהסימולטור, הפחתת שונות ואיבר אנטרופיה המחושב בדיוק. אנו מצדיקים את אומד הגרדיאנט ומשווים את תוחלתו לחישוב מדויק בבעיות קטנות. מסלול אחד לכל מדיניות נדגמת מספיק לתקינות האומד, אך חלוקת התקציב בין מדיניות שונות לבין מסלולים חוזרים משפיעה על הלמידה. בכל עדכון נדגמות מדיניות חדשות; בזמן ביצוע נשמרת המדיניות שנבחרה לכל אורך האפיזודה. אין בשיטה זו שלב של שכפול חלקיקים.

התכנון הניסויי מבחין בין תקינות הגרדיאנט, שגיאת משפחת הקירוב וביצועי הבקרה. תקציב האינטראקציות כולל גם את האתחול וגם את העדכון הווריאציוני. ההערכה מפרידה בין התחייבות למדיניות מלאה לבין הגרלת פעולה מחדש בכל ביקור. ניסויי פיתוח ראשוניים בסביבות רשת מצביעים על תועלת לדגימת יותר מדיניותות במקום חזרה מרובה על כל מדיניות, ועל ביצועים קרובים לבקרת PPO מסייעת בחקירה בניסוי בינוני אחד. בניסוי הקשה, הקצאה קטנה מדי לשלב החקירה לא יצרה מסלולים מוצלחים. חלוקה מחדש של אותו תקציב שיפרה את הביצוע, אך לא ביטלה את הפער בין כללי הביצוע. אלו תוצאות פיתוח, ולא הוכחה לעליונות כללית.

העבודה כוללת גם שיטות דגימה חלופיות, בדיקות תקינות ותיעוד של ניסיונות שלא הצליחו. השוואה מלאה מול שיטות הבסיס, חזרות בלתי תלויות, מפות שלא שימשו לפיתוח וניתוח שגיאת הקירוב עדיין נדרשים להשלמת המסקנות.



אוניברסיטת בן-גוריון בנגב

הפקולטה למדעי הטבע

המחלקה למדעי המחשב

בחירת מדיניות באמצעות הסקה לפי תוחלת התגמול

חיבור זה מהווה חלק מהדרישות לקבלת התואר מוסמך למדעי הטבע

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